1. Human Influences on Ecosystems

1.1. Food Supply

Humans have increased food production because:

- **Agricultural Machinery** to use larger areas of land and improve efficiency
- **Chemical Fertilisers** help crops grow better
- **Insecticides**: a type of pesticide that kills insects
- **Herbicides**: a type of pesticide that kills weeds
- **Selective Breeding** to improve production by crop plants and livestock

Large-scale Monoculture: the continuous production of one type of genetically identical crop.

- **Positive Impacts of Large-scale Monoculture**
  - Increasing yield due to single crop specialisation
  - Lower food prices
  - Reduce labour costs
- **Negative Impacts of Large-scale Monoculture**
  - If a natural disaster occurs, **the whole crop could be wiped out.**
  - If pests & diseases attack crops, they could harm them easily
  - Using large fields and pesticides reduces the variety of species. This hinders biodiversity.
  - When insecticides are used persistently, the pests may eventually become **resistant** to them, reducing their effectiveness
- **Positive impacts of Intensive Livestock Production**
  - Producing large quantities of food needs less space.
  - Controlled environments allow for year-round food production.
  - Lower production costs make it possible to offer food at more affordable prices.
- **Negative Impacts of Intensive Livestock Production**
  - Welfare issues for the livestock
  - Diseases can spread easily among them
  - Waste can pollute land and waterways nearby

1.2. Habitat Destruction

Biodiversity: the number of different species that live in an area.

- **Reason for habitat destruction**
  - Increased area for food crop growth, livestock production, and housing
  - Extraction of natural resources
  - Freshwater and Marine pollution
- By altering food webs and food chains, humans can harm habitats.
- **Effects of deforestation**
  - Reduced biodiversity/destroys habitats/extinction
  - Loss of CO_2 fixation, thus increase in CO_2, thus global warming
  - **Soil erosion:** tree roots cannot retain soil and go into rivers, making the water dirty & causing blockages, and the soil becomes less fertile
  - **Flooding:** 75% of water is usually absorbed by foliage, root systems or evaporates. After deforestation, water accumulates in valleys.

1.3. Pollution

Pollution due to pesticides:

- **Insecticides (kill insects):** meant to kill insects which eat crops, but can kill other useful insects such as bees, which are pollinators, or by bioaccumulation (the increase in the dose of toxin from one level of the food chain to the next)
- **Herbicides (kill weeds):** can be harmful to animals which eat the plants

Non-biodegradable plastics:

- Choke birds, fish and other animals
- Fill up the animals' stomachs so that they can't eat food
- Collect in rivers and get in the way of fish

Global Warming:

- Increase in the average temperature of the Earth
- Methane from the burping of cows
- It started at the same time as humans began burning fossil fuels
- Scientists have proven that fossil fuels are one of the main factors causing this
- It increased carbon dioxide and methane concentrations in the atmosphere, enhancing the greenhouse effect that leads to climate change.

Eutrophication: when water plants receive too many nutrients.

- Fertilisers are put in soil by farmers.
- Fertilisers with nitrates/detergents with phosphates leach into rivers and lakes after rain.
- Water plants grow more than usual
- They block sunlight and kill plants underneath
- They die and sink to the bottom
- Bacteria/fungi decompose remains using the O_2 and decreasing the O_2 concentration
- Fish and other creatures die from oxygen starvation

1.4. Conservation

Sustainable Resource: one which is produced as rapidly as it is removed from the environment so that it does not run out

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- Some resources can be conserved and managed sustainably but are limited to forests and fish stocks.
    1. **Forests** can be conserved using education, protected areas, quotas and replanting.
    2. **Fish stocks** can be conserved using education, closed seasons, protected areas, controlled net types and mesh size, quotas and monitoring.

Natural Resources:

- **Water:** used to grow food, keep it clean, provide power, control fires, and drink. We get water constantly through rainfall, but we use the planet's freshwater faster than it can be replenished.
- **Fossil fuels** must be conserved as they will soon run out; therefore, they should be replaced with green energy.

Recycling:

- **Water:** water from sewage can be returned to the environment for human use by sanitation and sewage treatment
- **Paper:** sent to special centres where it is pulped to make raw materials for industry
- **Plastic:** fossil fuels, bottles → fleece clothing
- **Metal:** mining takes a lot of energy, so recycling saves energy

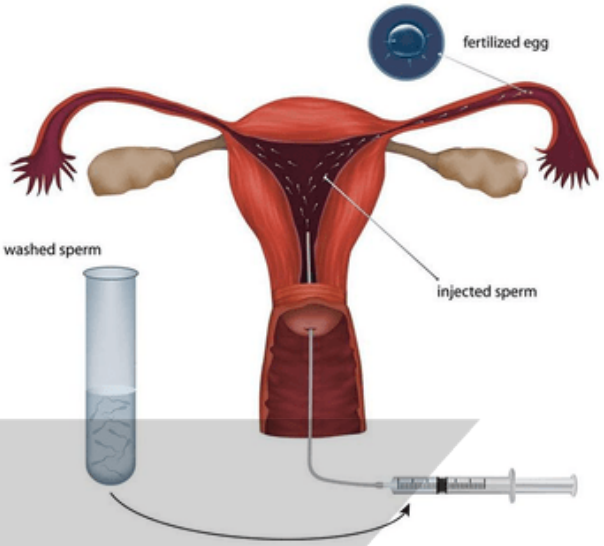
Species and habitats need to be conserved because:

- Organisms have value in themselves (ethical value)
- Maintaining biodiversity to reduce extinction
- Value to medicine (new molecules from exotic plants = new drugs)
- Genetic resources are helpful to humans as well and are lost when species disappear (DNA for genetic engineering)
- Each species has its role in its ecosystem; if it is removed, then the whole ecosystem could collapse

Endangered species:

- How they become endangered: climate change, habitat destruction, hunting, pollution and invasive species
- If the population size drops, variation decreases
- Endangered species can be conserved by monitoring and protecting species and habitats, education, captive breeding programmes, and seed banks
- Reasons for Conservation Programmes include:
    - reducing extinction
    - protecting vulnerable environments
    - maintaining ecosystem functions by nutrient cycling and resource provision, e.g. food, drugs, fuel and genes
    - increase biodiversity

1.5. Artificial Insemination in Captive Breeding Programmes



- The sperm/gamete is collected from the male.
- Males can be chosen from different breeding programmes to maximise genetic variation.
- The sperm is screened for quality checks.
- After checking, the sperm/semen is frozen, and the female is given drugs to induce ovulation.
- The sperm/semen is then placed in the vagina/uterus/oviduct, where the sperm/semen is inserted at the appropriate time for ovulation.

1.6. *in vitro* Fertilisation (IVF) in Captive Breeding Programmes

- It involves fertilisation of an egg outside the mother's body
- *Method*
    - Eggs are extracted from the ovaries of the mother using a needle
    - Eggs are kept in a culture medium.
    - Male semen is mixed with the eggs so fertilisation takes place
    - Zygotes form which develop into embryos
    - These embryos are checked after being left in the culture medium for several days
    - The embryos are then transferred into the mother or a surrogate.

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